

Music Listening Analytics and Recommendation System

Using PySpark and the Last.fm 1K listening dataset

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Project Scope

The submission covers the required music recommendation system and the extra analytics work for bonus marks

Part A

Big Data Analytics

Spark aggregations, temporal patterns, country analysis, and KMeans listener segmentation.

BONUS

Part B

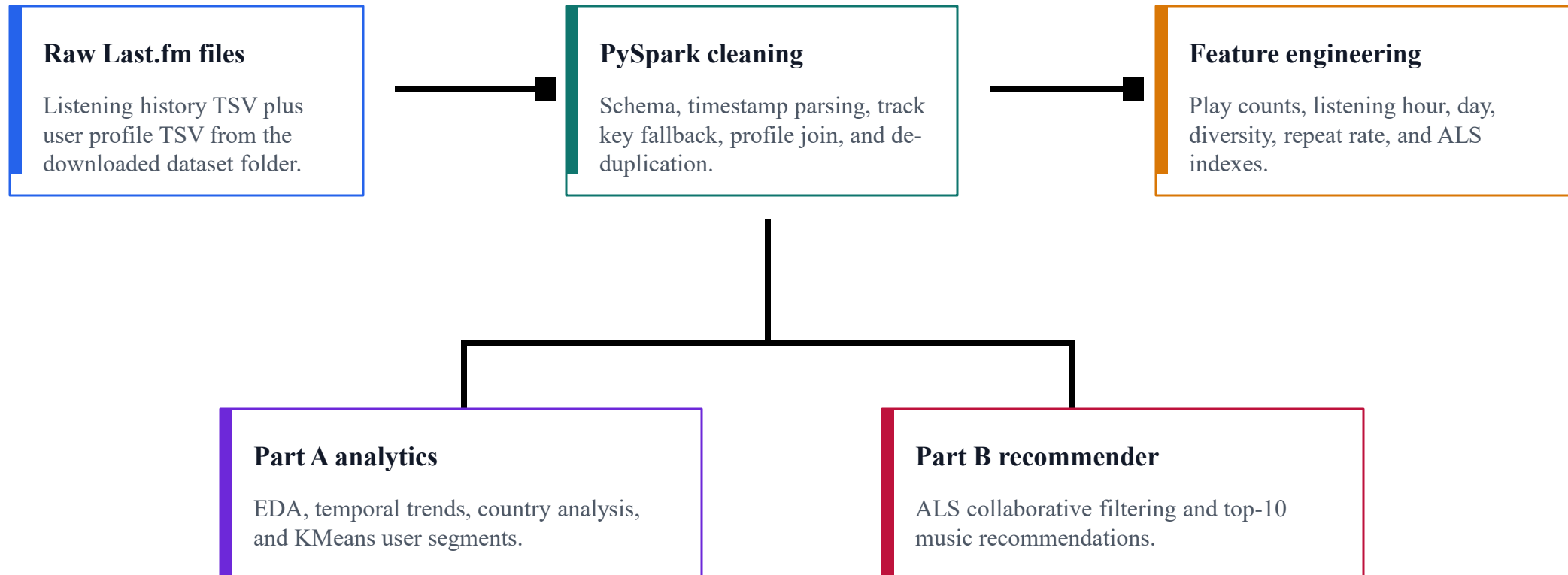
Recommendation System

Spark MLlib ALS on implicit feedback, model evaluation, and top-10 song recommendations.

Submission files: two executed Jupyter notebooks, README, generated CSV outputs, and this presentation.

Data Processing Workflow

The same cleaned Last.fm data is used for analytics, clustering, and music recommendation

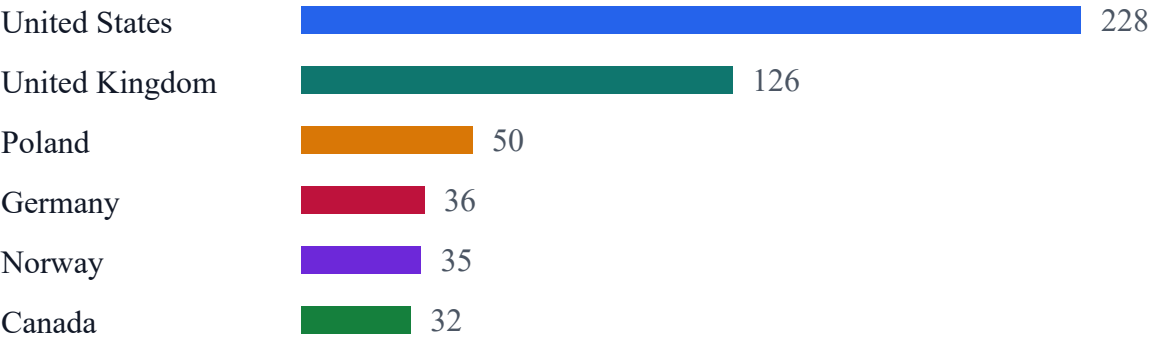


Dataset Summary

The notebook uses a laptop-safe sample of 1,000,000 listening events from the Last.fm 1K dataset



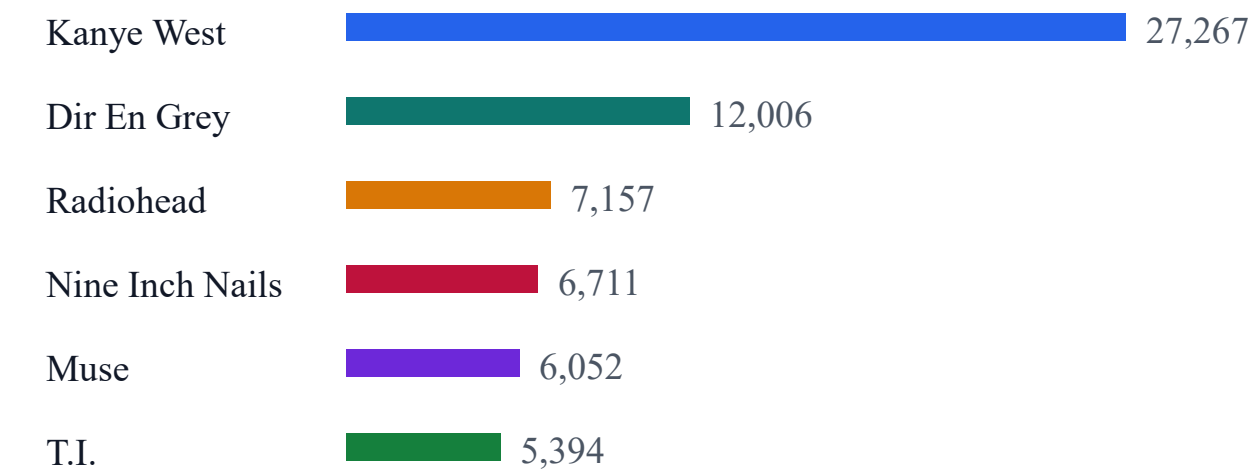
Top profile countries



This sample is large enough to show Spark loading, cleaning, joins, aggregations, feature preparation, and model training.

Big Data Analytics Results

Spark aggregations are used to find the most listened artists and tracks



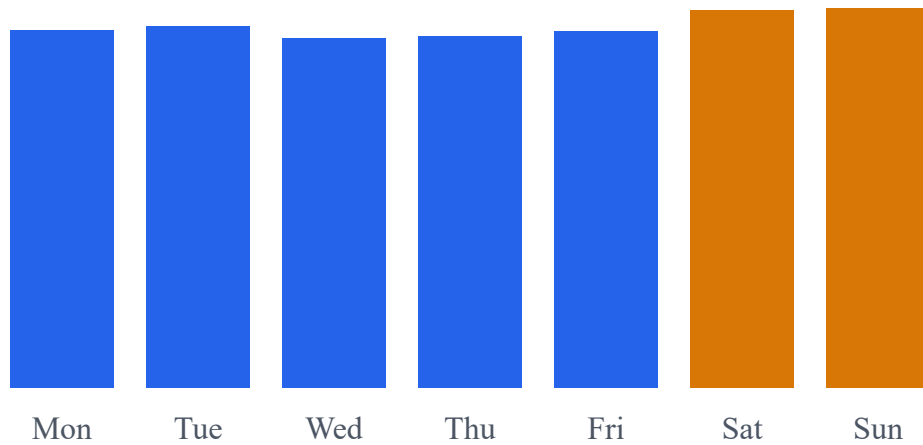
Top track	Artist	Events
See You In My Nightmares	Kanye West	2,069
Say You Will	Kanye West	2,065
Heartless	Kanye West	2,063
Love Lockdown	Kanye West	2,062

Finding: a small group of artists and tracks receives many repeated listens, so play count is useful as implicit feedback.

Listening Time Analysis

Timestamp features are used to compare listening activity by day and hour.

Listening events by day



Evening activity by hour



Result: Sunday has the highest listening volume, and 18:00 is the most active hour in the sample.

User Segmentation With KMeans

Spark ML groups listeners using volume, diversity, repeat behavior, and time-based features.

Segment	Users	Avg listens	Repeat rate	Label
0	9	56,323	0.8273	Heavy listeners
1	28	13,596	0.6861	Balanced listeners
2	4	28,098	0.5590	Music explorers

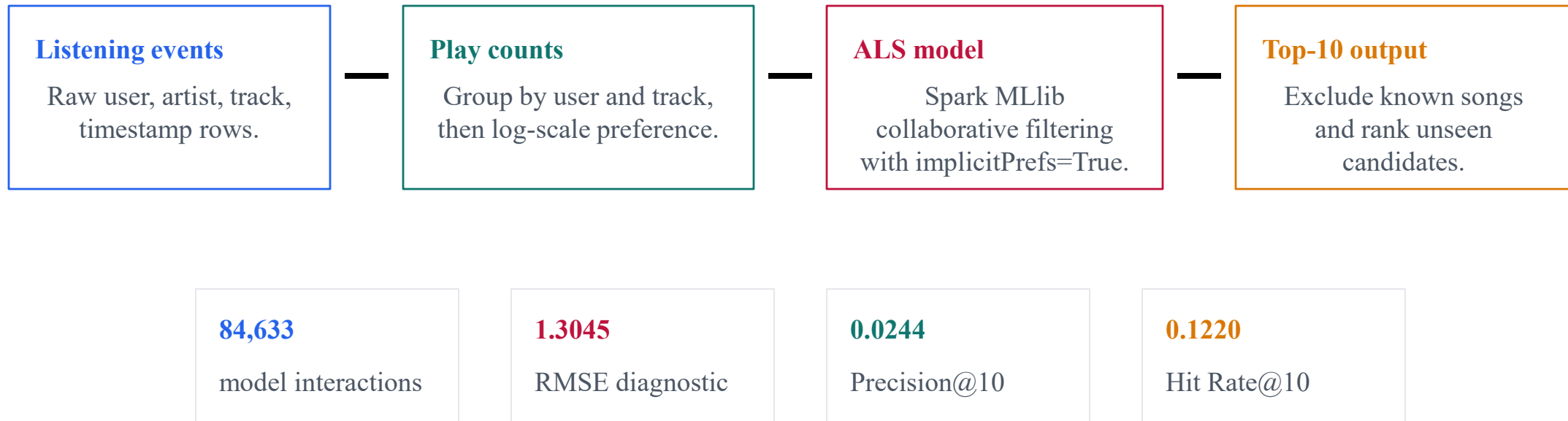
Feature set

- total listens
- unique artists
- unique tracks
- active days
- avg listens / day
- artist diversity
- repeat rate
- night share
- weekend share

Output: the users are grouped into heavy listeners, balanced listeners, and music explorers

Recommendation Model

Spark MLlib ALS uses listening counts as implicit feedback to produce ranked song recommendations



Recommendation Output

The model scores candidate tracks and removes songs already listened to by the selected user

Rank	Recommended track	Artist	Score
1	Stand Up Guy	T.I.	1.6746
2	Cream	Prince	1.4785
3	Me And Those Dreamin' Eyes Of Mine	D'Angelo	1.4744
4	Rain On Me	Ashanti	1.4671
5	Kingdom Of Doom	The Good, The Bad & The Queen	1.4627
6	Sweet Sweet	The Smashing Pumpkins	1.4592

Selected user: user_000031. The notebook generates top recommendations that can be shown during the demo.

Files Prepared for Submission

The project demonstrates big data analytics and a recommendation system using PySpark.

Item	File / evidence
Part A notebook	Music_Listening_Analytics_Bonus_PartA_PySpark.ipynb
Part B notebook	Music_Recommendation_PySpark.ipynb
Dataset link	Last.fm Dataset - 1K users
Outputs	CSV summaries, segments, metrics, and recommendations
Guide	README.md and BONUS_SUBMISSION_GUIDE.md

Demo flow: open Jupyter, run Part A analytics and clustering cells, then run Part B ALS training and recommendation output cells.